# Recreating Video Arenas via Automated Preference Scoring

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## Abstract

Evaluating video generation models is notoriously challenging. Existing metrics either fail to align with human judgment or rely on costly, unscalable human ratings. In this work, we introduce **Video Preference Score** (VPS), a fully automated video preference score designed to directly model human judgment and accurately recreate large-scale, Elo-based video arenas. Trained on a newly collected dataset that augments generated videos with retrieved real-world footage, VPS evaluates paired videos against complex text prompts to predict human preference. Experiments demonstrate that VPS correlates highly with human consensus and generalizes zero-shot across unseen models and benchmarks. Furthermore, we showcase the utility beyond evaluation: as a plug-in reward model, VPS seamlessly integrates with Flow-GRPO to improve WAN-2.1's generative quality by 51 Elo points. Overall, VPS provides a scalable and generalizable solution for both evaluating and aligning modern video generation models.

## 1 Introduction

Video generation models have achieved tremendous progress and have shown promising capabilities in visual reasoning [24, 31], world simulation [3, 36], and making real-world decisions [13, 22, 37]. However, evaluating video generation models remains difficult despite the availability of various evaluation metrics. Existing evaluation metrics fall mainly into two categories: *rule-based* and *rating-based*. Rule-based metrics [12, 15, 25] compose arithmetic operations from pre-trained visual features. These metrics are clearly defined, deterministic, and reproducible. However, many of them require reference input, which is difficult to obtain for open-world video generation. In addition, rule-based metrics often align poorly with human study, since visual perception is, by its nature, subjective. Rating-based metrics [10, 9, 35] instruct human annotators to assign a rating given certain criteria. However, human annotation is costly and time-consuming. The evaluation result is sensitive to the size and representativeness of the participants. Meanwhile, evaluation details are glossed over in most technical reports. As a result, human evaluations are often considered hard to replicate.

In this paper, instead of introducing a brand-new benchmark, we start from an online video arena leaderboard and ask the following question: *How well can we recreate a video arena with minimal effort?* Given a fixed set of text prompts and videos generated by several video models, we can rebuild an Elo-based leaderboard by rerunning video-to-video comparisons based on any independent or pairwise score. The correlation between the recreated leaderboard and the public leaderboard then tells us how well the used video score aligns with human judgments.

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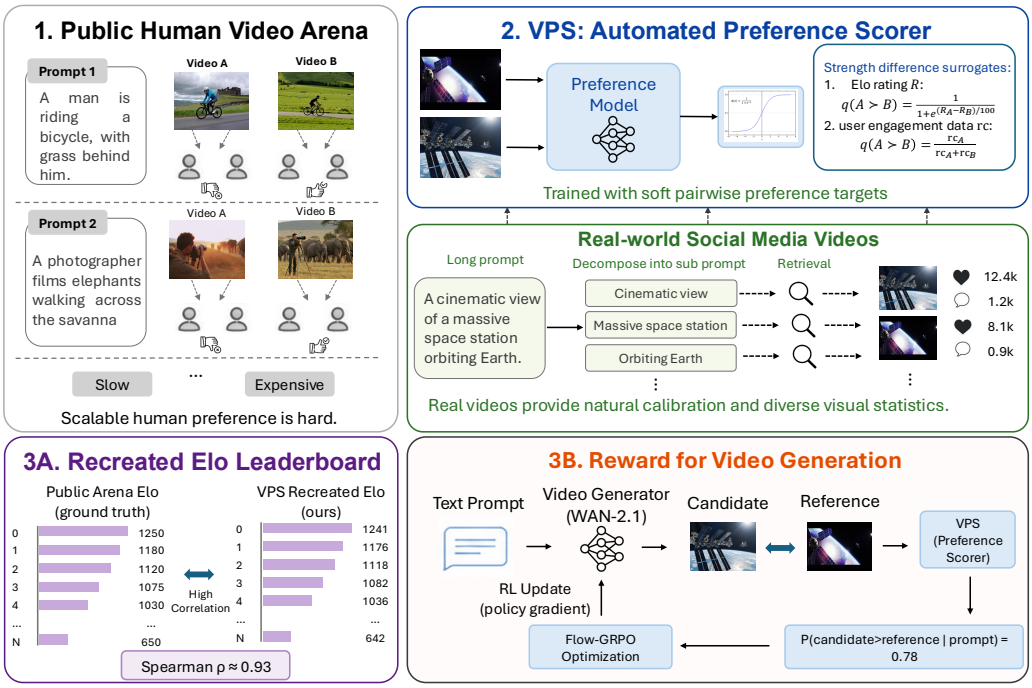



Figure 1: An overview. We propose VPS, an automated preference score to approximate public video arena based on large-scale human votes. We show that VPS is able to recreate the Elo-based leaderboard with high correlation and serve as a reward model for reinforcement fine-tuning a video generation model.

To this end, we propose a fully automatic evaluation system that measures alignment with the human preferences of a large crowd. We prioritize human preference and propose to model it directly. The proposed **video preference score**, or VPS for short, takes as input paired videos and a text prompt, and outputs a scalar value representing how likely one video is preferred over the other. To learn VPS, we construct a dataset with both generated videos from the up-to-date video generation models and real-world videos from social media. We propose a data collection pipeline to retrieve highly relevant videos given long, complicated text prompts, which are challenging but more commonly used in modern video evaluation benchmarks.

Figure 1 gives an overview of our approach. We summarize our key findings as follows. First, VPS aligns with human preference on video arena leaderboard, outperforming existing rule-based and rating-based metrics. Second, the learned human preference score generalizes across models and benchmarks. We show that the VPS learned from a subset of video generation models correctly predicts the preference order of unseen models with a correlation of 0.710. VPS learned from the Artificial Analysis leaderboard remains robust on Arena AI. Third, real-world videos serve as a good source to augment the video preference data. Relative social-media engagement between two real videos retrieved for the same prompt provides a weak but freely available preference signal, and adding social-video data improves downstream reinforcement fine-tuning even though it slightly weakens arena-only reconstruction. Finally, we show that the preference score serves as a plug-in reward model and improves the quality of video generation. Flow-GRPO with VPS shows a quantitative improvement of 51 Elo points. Qualitative results also support our findings.

## 2 Related Work

**Video Evaluation.** Video evaluation is multifaceted, and many evaluation metrics have been proposed. Fréchet Video Distance (FVD) [25] measures the distributional distance between the real-world data and the generative model, both of which are approximated with multivariate Gaussians from some pre-trained feature embeddings, following FID [11] in images. FVD, as a metric, is often criticized by

(1) an impractically large sample size required for reliable estimation [19], (2) an underrepresentative pretrained video encoder [4], and (3) bias towards static scenes [7]. ViCLIP [27], T2VScore [33], and ETVA [8] emphasize text-to-video alignment while ignoring other dimensions. Video Quality Assessment metrics [30, 32] mainly capture the aesthetic aspect of videos. More comprehensive benchmarks, such as VBench [12] and EvalCrafter [18], evaluate video generation quality in multiple dimensions by putting together the aforementioned metrics and others. However, their reliance on external pre-trained models increases evaluation costs and makes end-to-end optimization infeasible. Inspired by the success of open platform [5] to evaluate LLMs, crowd-sourced, voting-based video arenas have been built. The rankings are so far the most reliable and more widely recognized However, thousands of human votes are required for each model, making it hard to replicate. Our work confirms the evaluation consistency of these voting-based leaderboards and also proposes a purely computational approach to recreate the arena without repeated human effort.

**Human Preference Models.** Human preference is a simple and effective objective to solve complex tasks when the reward function is ill-defined [6, 1, 21]. Popularized in Large Language Models [41], human preference is later adopted to evaluate and improve image generation models, including PickScore [14], MPS [40], HPS [34, 20]. It is a natural extension to improve video generation with human feedback. VideoScore [10] proposes one of the first large datasets with human ratings for generative videos. VideoScore2 [9] further includes detailed chain-of-thought rationales. However, recent studies [17, 35] reveal that pointwise ratings do not align well with ranking-based human preference. In contrast, our VPS directly models pairwise preference, resulting in higher correlation. VideoReward [17] defines fine-grained multi-dimensional human preferences and learns a reward model by fine-tuning a vision language model. VisionReward [35] further increases the evaluation dimensions from 5 to 20. Although this fine-grained assessment improves interpretability, the multi-dimensional evaluation unavoidably introduces new complexity in determining the final preference from multiple axes.

## 3 Data Collection

We construct a pairwise preference dataset to learn automatic preference score. This video preference score can be used for reconstructing the video arena without laborious human votings and serve as a reward model for reinforced fine-tuning video models. In order to adhere to the arena-style ranking, we build the dataset with two main design goals. First, the dataset should be large enough to support arena-style model comparison without collecting a new round of human annotations. Second, the data should contain realistic visual statistics that are missing in most existing benchmarks which focus on generated videos only. To this end, we combine generated videos from a public video arena with real-world social-media videos retrieved for the same prompts. The resulting dataset supports three types of comparisons: generated–generated, real–real, and real–generated.

### 3.1 Prompts

We collect prompts from two existing sources. The first consists of short prompts similar to VBench [12], which typically describe a single scene or action. The second consists of longer paragraph-style prompts from modern video-generation benchmarks, in particular, the Artificial Analysis video arena[1]. These longer prompts are more representative of current text-to-video use cases: they often contain multiple entities, attributes, scene constraints, temporal relations, and stylistic requirements.

### 3.2 Generated videos from video arenas

We scrape generated videos from the Artificial Analysis video arena. For each example, we store the prompt, the two generated videos, the corresponding models that produced them, and the public Elo score of the corresponding models. These model-level Elo scores provide weak but scalable supervision: if two videos are generated by models with different Elo scores, the video from the higher-Elo model is treated as more likely to be preferred, with the strength of preference determined by the Elo gap rather than by a hard binary label. This lets us convert an existing public arena into a training set for pairwise preference prediction, the details of which are provided in the following

[1] `https://artificialanalysis.ai/video/arena`

section. The Artificial Analysis subset contains 7,372 generated videos corresponding to 694 different prompts. Videos generated under the same input prompt can be paired arbitrarily, resulting in 27,498 pairwise examples.

### 3.3 Real-world videos from social media

Generation-only datasets have an important limitation: their visual diversity and quality are limited by the set of models available at collection time. In contrast, real-world social-media videos contain natural camera motion, lighting, physical dynamics, compression artifacts, human activity, and long-tail scenes that are difficult for current generators to cover. We therefore augment arena data with real videos retrieved from social media.

We develop a social media retrieval API that takes a query string or a list of keywords, and returns relevant video URLs, metadata (such as the observed engagement with the video, including user likes and comments), and the timestamp of the segment judged most relevant to the query. The timestamp is important because full social-media videos often contain intros, transitions, or off-topic segments.

**Retrieval for short prompts.** For short VBench-style prompts, direct retrieval works reasonably well. A query such as "a cat jumps onto a couch" closely resembles the search queries used by social platforms, and the top returned videos are usually semantically tight.

**Long prompts require decomposition.** In contrast, Artificial Analysis-style prompts are longer and harder. A single paragraph-level query often contains several semantic components, and social media search may match only the most salient entity while ignoring the action, temporal constraint, or scene attribute. For example, a prompt describing "Machu Picchu at sunrise with mist rising from the valleys" may retrieve generic Machu Picchu videos while missing the sunrise or mist components.

To address this, we decompose each long prompt $\boldsymbol{y}$ into a set of shorter sub-prompts $\{q_i\}_{i=1}^{m}$. Each sub-prompt targets one major semantic component of the original prompt, such as an object, location, action, visual attribute, or temporal condition. We query the retrieval API independently with each sub-prompt and aggregate the returned videos.

Let $\mathcal{R}_i$ denote the ranked set of videos returned for sub-prompt $q_i$. We form the candidate pool $\mathcal{U}(\boldsymbol{y}) = \bigcup_{i=1}^{m} \mathcal{R}_i$. Each candidate video $v \in \mathcal{U}(\boldsymbol{y})$ is assigned an aggregation score $s_{\mathrm{ret}}(v; \boldsymbol{y}) = \sum_{i=1}^{m} \mathbf{1}[v \in \mathcal{R}_i] - \lambda \sum_{i:v\in\mathcal{R}_i} \mathrm{rank}_i(v)$, where the first term rewards videos retrieved by multiple sub-prompts and the second term breaks ties using the API-provided ranking. We then retain the top videos under $s_{\mathrm{ret}}$. This favors videos that match multiple components of the prompt rather than videos that only match a single keyword.

**Prompt decomposition improves real-world retrieval.** Table 1 reports a retrieval ablation. We use the FVD score between the set of retrieved videos and generated videos as a proxy to measure the retrieval quality. For short VBench-style prompts, using relevance ranking produces a lower FVD score than random retrieval. The improvement is more significant for long prompts, where top-ranked videos from chunked retrieval reduce FVD from 100 to 45 relative to random sampling from the retrieved pool. These results support our use of prompt decomposition and relevance-ranked aggregation when constructing real-world anchors for long-form video-generation prompts.

Table 1: Retrieval ablation. Prompt decomposition substantially improves real-world retrieval for long prompts. Lower FVD is better.

| Prompt set | decomposition | ranking | $\mathrm{FVD}_{\downarrow}$ |
|---|---|---|---|
| VBench | | | 60 |
| VBench | | ✓ | 54 |
| AA | ✓ | | 100 |
| AA | ✓ | ✓ | 45 |

### 3.4 Data split & statistics

**Data splits.** We have a total of 694 prompts, and split train/val/test at the prompt level by default: no prompt appears in more than one of train, validation, or test. The *prompt-based data split* prevents leakage through repeated prompts and near-duplicate prompt templates. In addition, we experiment with *model-based data split*, ensuring that videos generated by certain models are never seen during training. This is useful to see whether the learned preference score generalizes to newer unseen models, which are likely to be introduced constantly.

Table 2: Dataset statistics for prior pointwise- and pairwise-supervision resources and our three splits.

| Dataset | Cutoff date (MM/YY) | # prompts | # videos | # triplets |
|---|---|---|---|---|
| *Pointwise score* | | | | |
| VideoScore2 [9] | – | 2,933 | 27,168 | – |
| LiFT-HRA [29] | – | – | 10k | – |
| VideoFeedback [10] | – | 31.6k | 37.6k | – |
| *Pairwise score* | | | | |
| VideoGen-Reward [17] | – | 358 | 4,691 | 25,233 |
| VPS-AA (ours) | 02/26 | 694 | 7,372 | 27,498 |
| VPS-Social (ours) | 02/26 | 694 | 18,799 | 269,430 |
| VPS-Combined (ours) | 02/26 | 694 | 26,171 | 553,711 |

**Data statistics.** Table 2 shows the statistics of the proposed VPS dataset (live at huggingface). By including both generated and real videos, the entire VPS dataset is an order of magnitude larger than existing preference-based datasets, *e.g.*, VideoGen-Reward [17], and on par with the video datasets with pointwise scores. Compared with pointwise scores, we will shortly demonstrate its better alignment with human preference, which is, by definition, a pairwise measurement.

## 4 Learning Automated Video Preference Score

We learn a pairwise preference model, VPS, that predicts which of two videos is more likely to be preferred for a given text prompt. Given a prompt $\boldsymbol{y}$ and two videos $\boldsymbol{x}^A$ and $\boldsymbol{x}^B$, the model outputs $f_\theta(\boldsymbol{y}, \boldsymbol{x}^A, \boldsymbol{x}^B) \in [0, 1]$, which estimates the preference of model $A$ over model $B$, denoted by $P(A \succ B \mid \boldsymbol{y}, \boldsymbol{x}^A, \boldsymbol{x}^B)$. Unlike pointwise evaluators that assign an absolute quality score to a single video, VPS is trained directly on pairwise comparisons, matching the structure of human preference arenas.

### 4.1 Preference target construction

Each training example is a tuple $(\boldsymbol{y}, \boldsymbol{x}^A, \boldsymbol{x}^B, q)$, where $q$ is the soft target probability that video $A$ is preferred to video $B$. The target construction depends on the sources of the two videos:

$$q(A \succ B \mid \boldsymbol{y}, \boldsymbol{x}^A, \boldsymbol{x}^B) = \begin{cases} \sigma\left((R_A - R_B)/100\right), & A, B \in \mathcal{D}_{\text{gen}}, \\ 0.75, & A \in \mathcal{D}_{\text{real}},\ B \in \mathcal{D}_{\text{gen}}, \\ 0.25, & A \in \mathcal{D}_{\text{gen}},\ B \in \mathcal{D}_{\text{real}}, \\ \dfrac{\text{rc}_A}{\text{rc}_A + \text{rc}_B}, & A, B \in \mathcal{D}_{\text{real}}, \end{cases} \tag{1}$$

where $\sigma(z) = 1/(1 + \exp(-z))$ is the sigmoid function following the Bradley-Terry model [2]. For generated–generated pairs, $R_A$ and $R_B$ denote the public Elo scores of the models that generate videos $A$ and $B$. This converts model-level arena rankings into soft pairwise supervision: a larger Elo gap implies a more confident preference target, while close-performing models produce softer labels.

For real–real pairs, $\text{rc}_A$ and $\text{rc}_B$ denote the `respond_count` values returned by the social-media retrieval API. The `respond_count` is the total observed engagement with a video, including likes, comments, and other platform-level responses. We use relative response count as the preference signal: video $A$'s winning probability is its response count normalized by the total response count of the pair. Engagement is used *only* for real–real pairs, and *only* as a relative count between two videos retrieved for the same prompt. Engagement remains a noisy preference proxy, as response counts are influenced by creator popularity, trends, and recommendation algorithms. This is different from an Elo-score score because the distribution of the raw response counts is long-tailed, while its logarithm resembles a normal distribution as illustrated in Figure 6. If both response counts are zero, we assign $q = 0.5$.

For mixed real–generated pairs, we assign the social-media video a fixed winning probability of $0.75$. This encodes a strong but non-deterministic prior that a relevant real-world video is more likely

to be preferred than a generated sample, since it typically contains more natural motion, camera statistics, physical consistency, and visual realism. Under the Bradley–Terry parameterization of Eq. 1, $q = 0.75$ corresponds to an edge of roughly 110 Elo, rather than a deterministic win. We avoid assigning probability 1.0 because retrieval noise and prompt mismatch can make some real-world videos imperfect anchors. We report a sensitivity analysis over $q$ in Appendix D (Table 6). Among the tested values, $q = 0.75$ performs best, while $q = 0.5$, which removes the real-over-generated preference prior, performs below the no-reward baseline.

This target construction is intentionally weakly supervised. Model-level Elo scores do not perfectly determine the quality of every generated sample, and social-media engagement is not a pure measure of perceptual quality. However, both signals are available *at scale*. Our forthcoming experiments test whether a model trained from these noisy pairwise targets, created from AA scraped 694 prompts, recover held-out arena rankings and provide a useful reward signal.

### 4.2 Preference model

We discuss the design choice of the preference model $f_\theta(\cdot)$.

**Metric-only baselines.** We first consider a metric-only baseline. This baseline asks whether existing metrics contain enough information to recreate human preference once a learned calibration layer is added. For each video, we compute a vector of existing automatic video-evaluation scores, including text-to-video alignment scores (CLIP [23], SigLIP [38], and InternVideo2 [28]), perceptual and quality metrics (LPIPS [39], DOVER [32], and human ratings (VideoScore [10]). We concatenate the scores with normalization and let $\mathbf{m}^A$ and $\mathbf{m}^B$ denote the metric vectors for videos $A$ and $B$. We train linear logistic regression and MLP baselines on pairwise metric features.

**Proposed end-to-end preference model.** Instead of relying on hand-designed metric features, our main model learns the preference score directly from raw pixels with learned video and text representations. We encode each video using Qwen2.5-VL-7B-Instruct and encode the text prompt using `sentence-transformers/all-MiniLM-L6-v2`. For each pair, we concatenate the two video embeddings and the prompt embedding, followed by an MLP $g_\theta$ with hidden dimensions of $[8192,\ 4096,\ 2048,\ 1024,\ 512,\ 256,\ 64,\ 4]$.

$$f_\theta(\boldsymbol{y}, \boldsymbol{x}^A, \boldsymbol{x}^B) = \sigma\left(g_\theta\left([\mathbf{e}^A, \mathbf{e}^B, \mathbf{e}^y]\right)\right), \tag{2}$$

where $\mathbf{e}^A, \mathbf{e}^B \in \mathbb{R}^{3584}$ are the video embeddings and $\mathbf{e}^y \in \mathbb{R}^{384}$ is the prompt embedding. The ordering of the two videos is preserved, so the model directly predicts the probability that the first video wins.

**Training objective.** We train the model with soft binary cross-entropy:

$$\mathcal{L}(\theta) = -q \log f_\theta(\boldsymbol{y}, \boldsymbol{x}^A, \boldsymbol{x}^B) - (1-q) \log\left(1 - f_\theta(\boldsymbol{y}, \boldsymbol{x}^A, \boldsymbol{x}^B)\right). \tag{3}$$

For experiments on the combined dataset, which contains generated–generated, real–real, and real–generated pairs, we upweight AA–AA pairs by a factor of 10 relative to social and cross pairs to prevent the larger splits from overwhelming the smaller but arena-aligned AA split.

$$\mathcal{L}_{\text{weighted}}(\theta) = \frac{\sum_i w_i\, \ell\left(f_\theta(\boldsymbol{y}_i, \boldsymbol{x}_i^A, \boldsymbol{x}_i^B), q_i\right)}{\sum_i w_i}, \qquad w_i = \begin{cases} 10, & \boldsymbol{x}_i^A, \boldsymbol{x}_i^B \in \mathcal{D}_{\text{gen}}, \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

We train with AdamW using learning rate $10^{-5}$, weight decay $10^{-5}$, batch size 256, and dropout 0.35. Training runs for up to 1000 epochs with early stopping based on validation loss. We reduce the learning rate on a validation-loss plateau and keep the checkpoint with the best validation loss.

### 4.3 Evaluation: recreating arena leaderboards

Although VPS is trained on pairwise targets, our main evaluation is whether it can recreate arena-style model rankings. We therefore convert predicted pairwise probabilities into per-model Elo scores and compare the reconstructed leaderboard against the public Artificial Analysis leaderboard.

For a held-out AA–AA pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ generated by models $m_i$ and $m_j$, the preference model predicts

$$\hat{p}_{ij} = f_\theta(\boldsymbol{y}_{ij}, \boldsymbol{x}_i, \boldsymbol{x}_j), \tag{5}$$

where $\hat{p}_{ij}$ is the predicted probability that video $i$ is preferred to video $j$. We convert this probability into an estimated pairwise Elo gap by applying a scaled logit transform:

$$\widehat{\Delta}_{ij} = K \cdot \mathrm{logit}(\hat{p}_{ij}) = K \cdot \log \frac{\hat{p}_{ij}}{1 - \hat{p}_{ij}}. \tag{6}$$

This gives a signed estimate of the relative quality gap between the two generating models. We then recover per-model scores $\{\hat{R}_m\}$ by solving a least-squares system over all predicted pairwise gaps:

$$\min_{\{\hat{R}_m\}} \sum_{(i,j)} \left[ (\hat{R}_{m_i} - \hat{R}_{m_j}) - \widehat{\Delta}_{ij} \right]^2. \tag{7}$$

Since Elo scores are identifiable only up to an affine transformation, we align the reconstructed scores to the public reference leaderboard with an affine calibration before reporting correlations.

For preference models trained on different splits of the dataset, we first infer the predicted preference score on the AA–AA pairs. This gives a consistent evaluation protocol across all training sources. We evaluate the reconstructed leaderboard using Spearman's $\rho$, Kendall's $\tau$, top-5 overlap, top-10 overlap, and median absolute rank error against the public leaderboards, including Artificial Analysis and Arena.ai.

### 4.4 Application: VPS as a reward model

Beyond evaluation, VPS can be used as a plug-in reward model to improve a video generator. Although VPS is trained as a pairwise preference predictor, we convert it into a scalar reward by comparing each generated video against a fixed prompt-matched reference. This yields a stable pointwise reward while preserving the pairwise semantics of the preference model.

We instantiate this reward in Flow-GRPO [16] to fine-tune WAN-2.1 [26]. For each training prompt $\boldsymbol{y}$, we precompute a reference video embedding $\mathbf{e}^{\mathrm{ref}}(\boldsymbol{y})$ from a base WAN-2.1 sample and a prompt embedding $\mathbf{e}^{y}$ from a sentence encoder. During RL training, the current policy generates a video $\boldsymbol{x} \sim \pi_\phi(\cdot \mid \boldsymbol{y})$. We encode $\boldsymbol{x}$ with the same video encoder used by VPS to obtain $\mathbf{e}(\boldsymbol{x})$, and define the reward as

$$r_\theta(\boldsymbol{y}, \boldsymbol{x}) = f_\theta\big(\boldsymbol{y}, \boldsymbol{x}, \boldsymbol{x}^{\mathrm{ref}}(\boldsymbol{y})\big) = \sigma\left(g_\theta\big(\big[\mathbf{e}(\boldsymbol{x}), \mathbf{e}^{\mathrm{ref}}(\boldsymbol{y}), \mathbf{e}^{y}\big]\big)\right), \tag{8}$$

where $g_\theta$ denotes the trained MLP preference head in Eq. 2. Intuitively, the reward is high when the generated video is predicted to be preferred over the prompt-matched reference video, and low when it is predicted to lose this comparison. This construction has two practical advantages. First, it avoids running pairwise comparisons against a large reference pool during RL, only one cached reference embedding per prompt is needed. Second, because the reference is fixed for each prompt, the reward remains comparable across multiple samples from the same prompt, which is important for group-relative policy optimization.

The finetuning experiments with WAN-2.1 [26] use Eq. 8 as the only external reward and a KL regularization to prevent the updated policy from drifting too far from the reference generator. The resulting optimization objective is therefore to maximize the VPS preference reward while constraining the updated generator to remain close to the reference policy. This demonstrates that VPS is not only an automatic evaluator for recreating arena rankings, but also a reward signal that can directly improve video generation quality.

## 5 Experiments and Findings

We evaluate VPS along three axes: (1) whether existing automatic metrics are sufficient to recover human preferences, (2) whether the learned preference model can recreate arena leaderboards, and (3) whether the learned score can serve as a reward for improving a video generator.

### 5.1 Existing metrics are useful but insufficient

We first evaluate how much existing automatic metrics align with human preference. We run

Table 3: Held-out text-to-video leaderboard reconstruction on AA pairs.

| Method | Spearman | Kendall $\tau$ |
|---|---|---|
| Metric linear reg. | 0.590 | 0.443 |
| VideoScore2 linear reg. | 0.628 | 0.456 |
| Metric MLP | 0.650 | 0.465 |

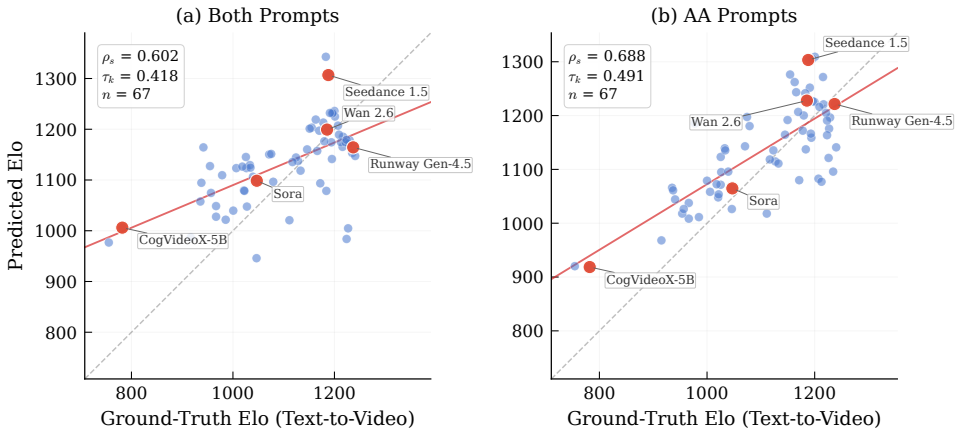



Figure 2: Recreated Elo scores versus public Artificial Analysis Elo scores. VPS-AA is specialized for AA-only leaderboard reconstruction, while VPS-Combined learns from both arena pairs and real-world social-video anchors.

linear logistic regression and MLP regression to predict pairwise preference and then evaluate by recreating the video arena leaderboard against the public Artificial Analysis (AA) leaderboard. Table 3 shows that existing metrics contain substantial preference signal. The metric MLP outperforms linear regression, indicating that nonlinear combinations of semantic, perceptual, and quality metrics are useful to approximate human arena judgments. However, VPS trained on only AA pairs (VPS-AA) achieves the strongest held-out rank correlation, with Spearman correlation $0.688$ and Kendall's $\tau = 0.491$. This suggests that learned prompt-conditioned video representations capture preference information that is not fully represented by fixed metric features. VPS trained on the mixture data (VPS-Combined) slightly underperforms. However, we will shortly show that VPS-Combined's inclusion of real-world videos benefits when VPS is used as a reward model for improving generation.

### 5.2 VPS recreates video arena and generalizes across leaderboards and models

**VPS generalizes to other leaderboards.** To test whether VPS-Combined learns a preference signal beyond the Artificial Analysis leaderboard, we evaluate it on the text-to-video leaderboard from Arena AI[2]. This setting provides an external leaderboard with a different model pool and ranking source. We consider the 31 models that are common to both AA and Arena AI leaderboards. Figure 3 (a-b) shows that VPS-Combined transfers well to this external ranking. On the overlapping model set, the reconstructed leaderboard achieves a Spearman correlation of $0.74$ and Kendall's $\tau$ of $0.57$ with the Arena AI leaderboard. These results suggest that VPS learns a preference signal that generalizes beyond the Artificial Analysis leaderboard.

**VPS generalizes to unseen models.** A practical evaluator should not only reproduce rankings over models seen during training but also rank newly released models as they appear. To test this, we separate a set of strong text-to-video models from training and use VPS to predict their Elo scores in the reconstructed leaderboard. This simulates a streaming evaluation scenario in which new, improved generators are introduced after the reward model has been trained. Figure 3 (c) shows the reconstructed Elo scores against the original Artificial Analysis Elo scores on the text-to-video test split. The held-out top-tier models are highlighted separately. Despite not being included during training, these models are assigned high reconstructed Elo scores, which is consistent with their ground-truth Elo score. However, it is worth noting that VPS tends to increase the performance gap.

This result is important for practical deployment. Video-generation leaderboards are dynamic: new models are frequently released, and rerunning large-scale human preference studies for every

[2] `https://arena.ai/leaderboard/text-to-video`

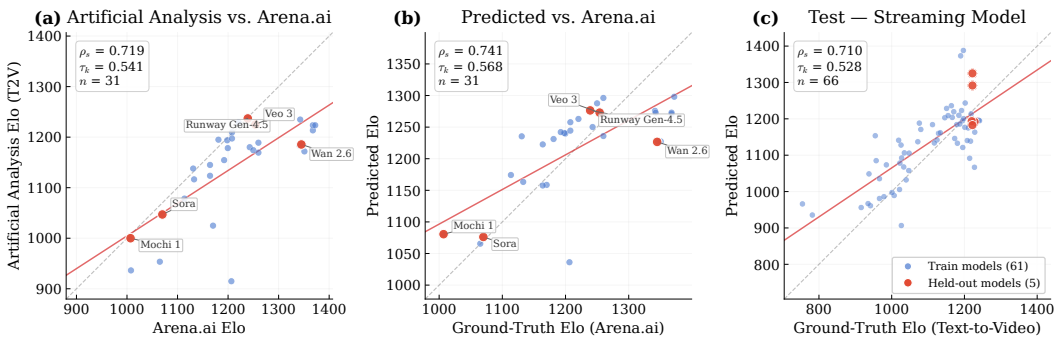



Figure 3: **(a-b)** Generalization to the Arena AI text-to-video leaderboard. VPS-Combined achieves Spearman correlation $0.74$ and Kendall's $\tau$ $0.57$, indicating that the learned preference model transfers beyond the Artificial Analysis leaderboard used during training. **(c)** Ranking held-out top-tier text-to-video models. Red circles denote strong models excluded from VPS training and evaluated only at test time. These held-out models are still reconstructed near the top of the leaderboard, suggesting that VPS can rank newly introduced high-performing generators without retraining on their arena comparisons.

model update is expensive. By accurately placing held-out high-performing models among the top reconstructed ranks, VPS provides evidence that it can support efficient leaderboard updates for newly released video generators.

**Agreement with sample-level human preferences.** We additionally evaluate VPS-AA zero-shot on MonetBench [35], a human-annotated pairwise preference benchmark. VPS-AA achieves 58.5% pairwise accuracy, compared with 59.5% for VQAScore and 54.9% for VideoScore. This provides sample-level evidence that the learned preference signal transfers to independent human judgments beyond leaderboard reconstruction. Full results are provided in Appendix D.

### 5.3 VPS provides a useful reward signal for video generation

Finally, we present the effectiveness of VPS as a reward model. The formulation follows Section 4. As reported in Table 4, both learned VPS improve video generation as a reward model. VPS-AA improves WAN-2.1 by $+9$ Elo, while VPS trained with both arena and social videos improves WAN-2.1 by $+51$ Elo. Thus, although adding social-media videos worsens AA-only leaderboard reconstruction, it substantially improves the downstream RL finetuning performance. This suggests that real-world videos provide complementary supervision beyond generated videos. Evaluation with VideoScore2 also supports the downstream gains: VPS-Combined achieves 2.683, compared with 2.526 for the base generator, 2.471 with random reward, and 2.577 with VideoScore2 reward (Table 7). Figure 4 shows that the predicted Elo score increases consistently with VPS-Combined while quickly saturating with VPS-AA. We show qualitative comparison in Figure 5 that WAN-2.1 with VPS indeed improves the generation quality in terms of camera motion, content relevance, and object movement. A multi-dimensional VBench evaluation shows improved multiple-object handling but a lower scene score (Table 8), indicating that the improvements vary across quality dimensions. More results are shown in Appendix G.

Table 4: Using VPS as a Flow-GRPO reward improves WAN-2.1. While VPS-AA is more specialized for AA-only leaderboard reconstruction, VPS-Combined provides the stronger reward signal for generator improvement.

| Reward model | Elo score | |
|---|---|---|
| | Baseline | Flow-GRPO |
| VPS-AA | 937 | 946 (+9) |
| VPS-Combined | 1066 | 1117 **(+51)** |

## 6 Conclusions

We propose VPS, a fully automated video preference score learned from weakly supervised signals including model-level Elo scores and user engagement on video-sharing platforms. VPS aligns well with human preference and shows robustness to unseen models and benchmarks. When used as a

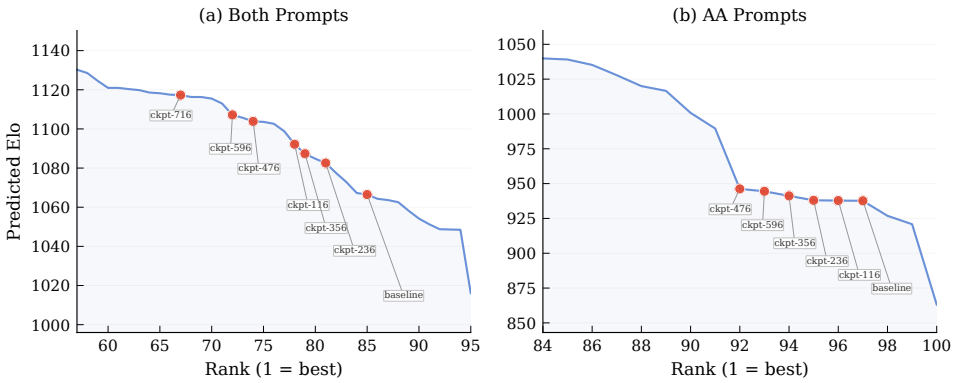



Figure 4: VPS-AA vs VPS-Combined as a reward model for video generation.

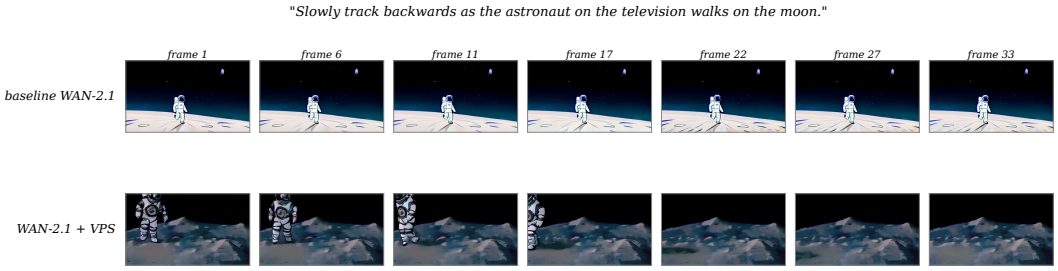



Figure 5: The video generated by WAN-2.1 after reinforcement fine-tuning shows enhanced motion and better alignment with text ("slowly track backward"). We show more results in Appendix G.

reward model, VPS greatly improves video generation quality. We hope this work will inspire more accurate and efficient video evaluation protocols.

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# A Limitations

We discuss potential limitations as follows.

**Weak supervision and imperfect proxies.** Both Elo scores at the model level from the video leaderboard and user engagements from the social media provide only noisy signals. They do not perfectly capture intrinsic perceptual quality. Therefore, the results should be interpreted as an empirical finding rather than a claim that social engagement alone is sufficient for true human preference alignment. The proposed VPS serves as a computationally efficient alternative.

**Evaluation precision.** Although VPS retains global arena rankings with high correlation, it demonstrates imperfect reconstruction at the very top of the leaderboards, reflected in a lower top-$K$ overlap metric. This indicates that recovering the exact order of top-performing models is more challenging than recovering the global trend. This is also probably due to the relatively small gap between the top-performing models nowadays.

**Social biases.** User engagement can be skewed due to other factors, such as trends, individual influencers, and recommendation algorithms, all of which could potentially result in social biases.

# B Societal Impacts

## B.1 Potential Positive Societal Impacts

**Democratized Evaluation.** By replacing costly, slow, and hard-to-replicate human rating arenas with a fully automated, highly correlated metric, VPS significantly lowers the barrier to entry for evaluating video generation models. This allows smaller academic labs and independent researchers to rapidly test and iterate on their models without the need for large-scale human annotation.

**Improved Model Alignment.** As demonstrated by its use as a reward model with Flow-GRPO, VPS provides a scalable feedback mechanism to improve the visual quality and prompt alignment of generative videos. This leads to more capable tools for creative professionals, educators, and content creators who rely on text-to-video technologies.

**Reproducibility in Research.** Human evaluations are notoriously sensitive to participant demographics, fatigue, and interface design. VPS provides a deterministic, computational framework for leaderboard recreation, fostering better reproducibility and standardized benchmarking within the generative AI community.

## B.2 Potential Negative Societal Impacts & Risks

**Amplified Social Media Bias.** A core innovation of VPS is the utilization of social media engagement (e.g., likes, comments, and response counts) as a proxy for human preference. However, social media engagement is heavily influenced by algorithmic recommendation biases. Training a reward model on this data risks implicitly teaching video generators to prefer sensationalist or biased visual styles over objective realism or diverse representation.

# C Safeguards

We implemented automated safety filters to remove NSFW or highly graphic content from the scraped social media videos prior to inclusion in the dataset. Furthermore, the dataset and pre-trained VPS reward model will be gated behind a Data Use Agreement (DUA) that requires researchers to verify their academic affiliation and agree not to use the dataset and the pre-trained reward model to optimize generators for explicit, deceptive, or non-consensual synthetic media.

# D Additional Evaluation

**Sample-level human preference agreement.** We evaluate VPS-AA zero-shot on MonetBench [35], a human-annotated pairwise preference benchmark. As shown in Table 5, VPS-AA achieves 58.5% pairwise accuracy, compared with 59.5% for VQAScore and 54.9% for VideoScore. Despite being trained for arena-style holistic preference, VPS-AA transfers to MonetBench's fine-grained, multi-dimensional preference criteria. VisionReward is reported as an in-distribution reference.

Table 5: Pairwise accuracy on MonetBench. VisionReward is evaluated in distribution, while the other methods are evaluated zero-shot.

| Method | Evaluation setting | Accuracy (%) |
|---|---|---|
| VQAScore | Zero-shot | 59.5 |
| VideoScore | Zero-shot | 54.9 |
| VPS-AA | Zero-shot | 58.5 |
| VisionReward | In-distribution | 72.1 |

**Sensitivity to the real–generated prior.** We sweep the real-over-generated preference prior over $q \in \{0.5, 0.75, 0.85\}$ and evaluate each resulting reward's effect on the fine-tuned generator with VideoScore2. Table 6 shows that $q = 0.75$ performs best among the tested values, reaching 2.664. The stronger prior $q = 0.85$ also improves over the no-reward baseline, whereas $q = 0.5$, which removes the real-over-generated preference prior, falls below it. These results support $q = 0.75$ as an empirical operating point.

Table 6: Sensitivity to the real-over-generated preference prior, evaluated through the resulting fine-tuned generator using VideoScore2.

| Prior | VideoScore2 |
|---|---|
| No-reward baseline | 2.526 |
| $q = 0.50$ | 2.465 |
| $q = 0.75$ | **2.664** |
| $q = 0.85$ | 2.577 |

**Comparison of reinforcement fine-tuning rewards.** We compare Flow-GRPO fine-tuning of WAN-2.1 using different rewards under identical settings and evaluate the resulting generators with VideoScore2. As shown in Table 7, random reward degrades the score relative to the base generator, while VPS-Combined achieves the highest score among the reported rewards. This supports the usefulness of VPS as a reward beyond simply applying reinforcement fine-tuning.

Table 7: Comparison of Flow-GRPO rewards. The resulting generators are evaluated with VideoScore2.

| Reward | VideoScore2 |
|---|---|
| No-reward baseline | 2.526 |
| Random | 2.471 |
| VideoScore2 | 2.577 |
| VPS-Combined | **2.683** |

**Multi-dimensional generation quality.** We evaluate the generator fine-tuned with VPS against the base generator on multiple VBench [12] dimensions. Table 8 reports the score changes and their 95% confidence intervals. Multiple-object handling improves, while the scene score decreases. The human-action score increases by 1.33 and appearance style changes by $-0.05$. These results complement the qualitative motion and alignment improvements while revealing a regression in scene quality.

## E Distribution of User Engagements

In Figure 6, we show that the raw response counts distribution is in a long-tailed distribution. After we take the logarithm, the distribution resemble a normal distribution, which is the Elo score prior, more closely.

Table 8: Changes in VBench scores after reinforcement fine-tuning with VPS, relative to the base generator. The average covers the four reported dimensions.

| Dimension | Change | 95% confidence interval |
|---|---|---|
| Multiple objects | $+4.37$ | $[+0.15, +8.69]$ |
| Human action | $+1.33$ | $[-4.00, +6.67]$ |
| Appearance style | $-0.05$ | $[-0.24, +0.15]$ |
| Scene | $-3.80$ | $[-6.73, -0.99]$ |
| Average | $+0.46$ | — |

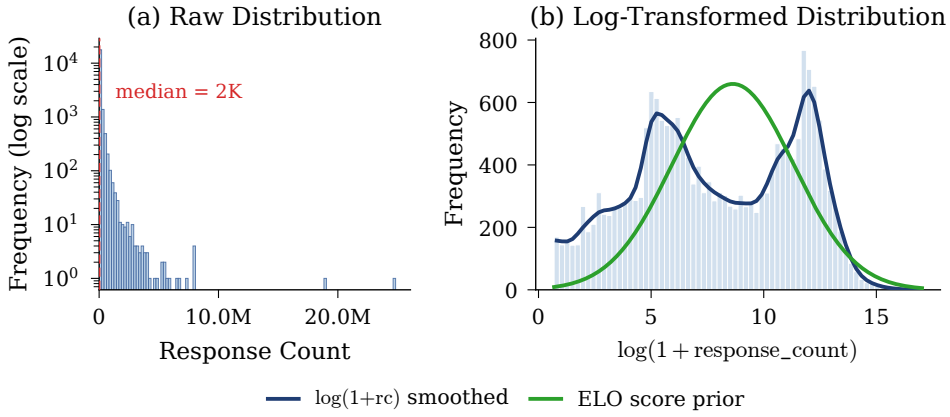



Figure 6: Respond count distribution

## F Example Training Pairs

Figure 7 illustrates how the training pairs for VPS are created.

## G More Qualitative Results

Figure 8 illustrates some examples of how VPS improves the quality of videos produced by WAN2.1.

## H Compute Resource

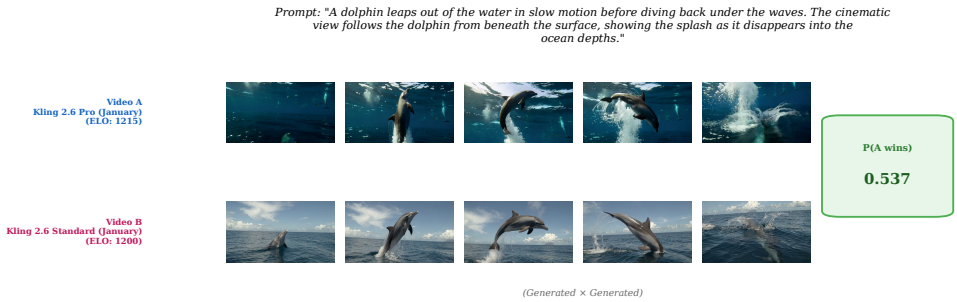
The preference model is trained on an internal cluster with NVIDIA L40s GPUs and takes around 15 minutes to train.

The Flow-GRPO experiments are conducted on an internal cluster with NVIDIA H100 GPUs due to a larger GPU memory requirement. Each experiment runs on $2\times$ H100 GPUs for 3 days, amounting to $\sim$ 150 GPU-hours per job.

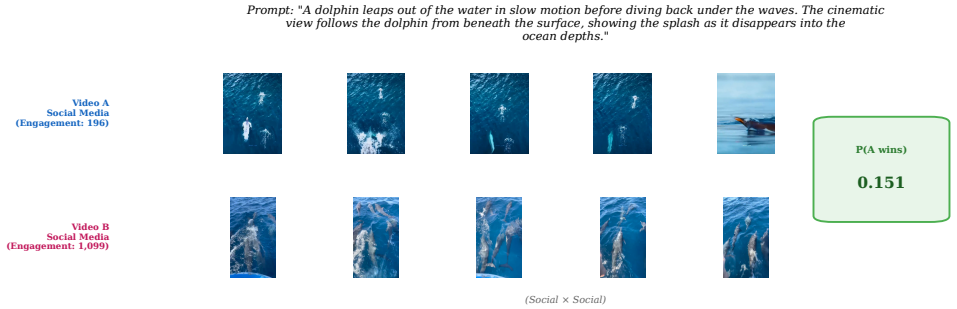
Preliminary and failed experiments do not take a significant amount of computation.

## I Licenses for existing assets

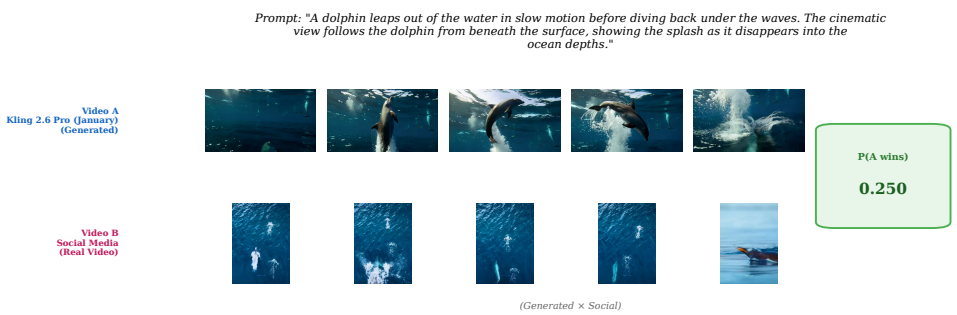
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(a) training pair from two generated videos

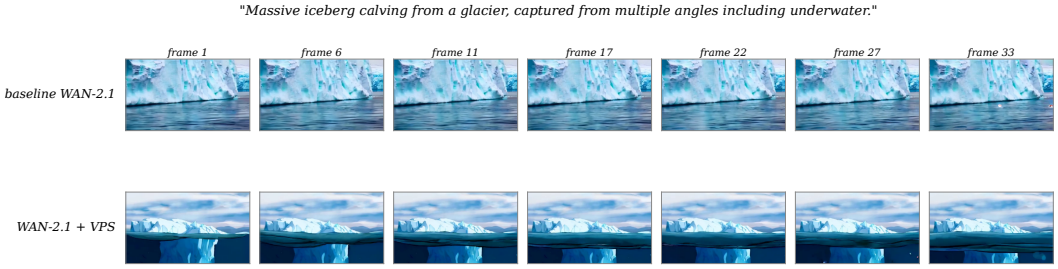



(b) training pair from two real world social media videos

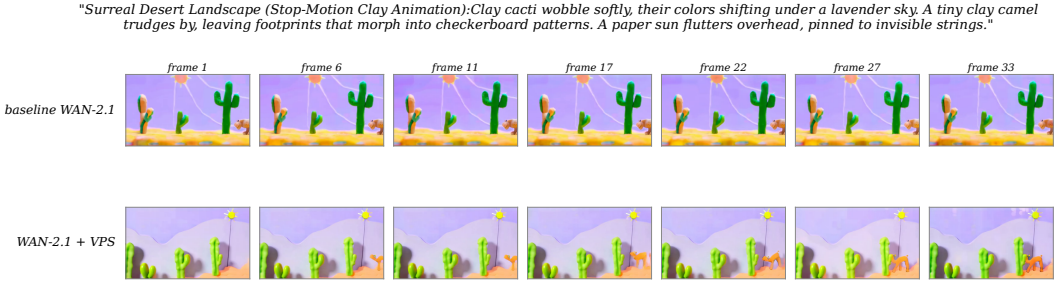



(c) training pair from a generated and a social media video

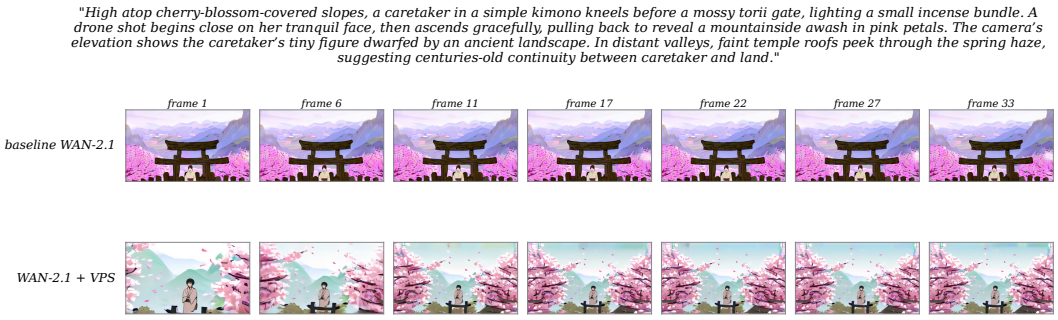
Figure 7: Example training pairs with assigned preference score



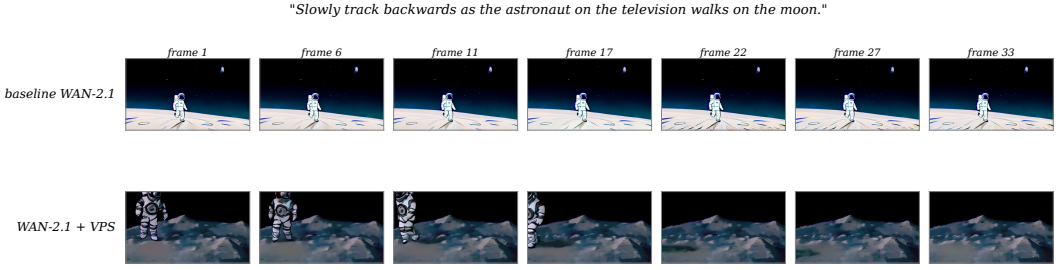

(a) baseline model fails to capture the iceberg underwater




(b) baseline model fails to capture the movement of the camel




(c) As per the prompt the camera is supposed to zoom out, but baseline model fails to capture this




(d) baseline model fails to capture the movement of the astronaut

Figure 8: VPS improves the alignment and hence the human preference over videos generated by WAN-2.1.

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